

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1229

Antelope Mine 138 kV
October 18, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1229. The Interconnection Customer proposes a 63.4 MW solar photovoltaic facility interconnecting at the Antelope Mine 138 kV substation in the West Virginia area. The request was received on June 18, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 413 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$20,765,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: AIR FORCE BASE 345-138 kV Transformer
Contingency: TIGER BAY-AIR FORCE BASE 345 KV
Loading: 121.9% (Rating: 541 MVA)
DFAX: 5.6%
Pre-project loading: 121.3%
2. Facility: NIPETOWN-DAM NO. 5 HYDRO STATION 138 KV
Contingency: AMERICAN-URBAN 345 KV
Loading: 121.7% (Rating: 314 MVA)
DFAX: 15.5%
Pre-project loading: 118.6%
3. Facility: CLARK COUNTY 345/138kV Auto
Contingency: ARTESIA COUNTRY CLUB-CLARK COUNTY 345 KV
Loading: 118.7% (Rating: 512 MVA)
DFAX: 6.1%
Pre-project loading: 118.0%
4. Facility: LASCASSAS-DAM NO. 5 HYDRO STATION 138 KV
Contingency: AMERICAN-URBAN 345 KV
Loading: 114.4% (Rating: 347 MVA)
DFAX: 15.5%
Pre-project loading: 111.6%
5. Facility: DAM NO. 4 HYDRO STA-SCHAFRATH SWITCH 138 KV
Contingency: AMERICAN-URBAN 345 KV Line
Loading: 113.4% (Rating: 317 MVA)
DFAX: 15.5%
Pre-project loading: 110.3%

6. Facility: CATAWBA ISLAND-TENGEN 138 KV Line
Contingency: AMERICAN-URBAN 345 KV line
Loading: 113.1% (Rating: 256 MVA)
DFAX: 18.4%
Pre-project loading: 108.5%
7. Facility: OFFERMAN SUBSTATION-CATAWBA ISLAND 138 KV line
Contingency: AMERICAN-URBAN 345 KV L
Loading: 109.1% (Rating: 254 MVA)
DFAX: 18.4%
Pre-project loading: 104.6%
8. Facility: ANITA-OFFERMAN 138 KV
Contingency: URBAN-AMERICAN 345 KV Line
Loading: 107.8% (Rating: 245 MVA)
DFAX: 18.4%
Pre-project loading: 103.0%
9. Facility: SCHAFRATH SWITCH-NIPETOWN 138 KV Line
Contingency: AMERICAN-URBAN 345 KV line
Loading: 105.3% (Rating: 354 MVA)
DFAX: 15.5%
Pre-project loading: 102.5%
10. Facility: CROCKETT-LORETTO 138 KV L
Contingency: AMERICAN-URBAN 345 KV
Loading: 104.2% (Rating: 578 MVA)
DFAX: 12.5%
Pre-project loading: 102.9%
11. Facility: DINOSAUR-LORETTO SUBSTATION 138 KV
Contingency: URBAN-AMERICAN SUBSTATION 345 KV Line
Loading: 103.8% (Rating: 300 MVA)
DFAX: 16.0%
Pre-project loading: 100.4%
12. Facility: SINTEK-ANITA 138 KV
Contingency: URBAN SUBSTATION-AMERICAN 345 KV line
Loading: 103.6% (Rating: 241 MVA)
DFAX: 18.4%
Pre-project loading: 98.8%
13. Facility: ANTELOPE MINE-PLANT 31 PAPER MILL 138 KV
Contingency: AMERICAN-URBAN SUBSTATION 345 KV
Loading: 101.3% (Rating: 308 MVA)
DFAX: 18.4%
Pre-project loading: 97.5%
14. Facility: HEALDSBURG-SINTEK 138 KV Line
Contingency: AMERICAN-URBAN 345 KV L
Loading: 100.5% (Rating: 237 MVA)
DFAX: 18.4%
Pre-project loading: 95.5%
15. Facility: ARTESIA COUNTRY CLUB 138/1 kV TX
Contingency: ARTESIA COUNTRY CLUB-AIR FORCE BASE 345 KV L
Loading: 100.2% (Rating: 688 MVA)
DFAX: 6.5%
Pre-project loading: 99.6%
16. Facility: ARTESIA COUNTRY CLUB 345 kV/1 kV Autotransformer
Contingency: ARTESIA COUNTRY CLUB-AIR FORCE BASE 345 KV

Loading: 100.2% (Rating: 688 MVA)
 DFAX: 6.5%
 Pre-project loading: 99.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
AIR FORCE BASE 345-138 kV Transformer	Replace transformer with higher-rated unit	\$18,849,000	3% (\$578,000)
NIPETOWN-DAM NO. 5 HYDRO STATION 138 KV	Rebuild line with higher-capacity conductor	\$1,748,000	51% (\$893,000)
CLARK COUNTY 345/138kV Autotransformer	Replace transformer with higher-rated unit	\$20,535,000	4% (\$733,000)
LASCASSAS-DAM NO. 5 HYDRO STATION 138 KV	Rebuild line with higher-capacity conductor	\$2,558,000	51% (\$1,307,000)
DAM NO. 4 HYDRO STA-SCHAFRATH SWITCH 138 KV	Rebuild line with higher-capacity conductor	\$2,136,000	51% (\$1,092,000)
CATAWBA ISLAND-TENGEN 138 KV	Rebuild line with higher-capacity conductor	\$9,328,000	71% (\$6,627,000)
OFFERMAN SUBSTATION-CATAWBA ISLAND 138 KV line	Reconductor line	\$1,839,000	71% (\$1,306,000)
ANITA-OFFERMAN 138 KV	Reconductor line	\$997,000	71% (\$708,000)
SCHAFRATH SWITCH-NIPETOWN 138 KV Line	Reconductor line	\$629,000	51% (\$322,000)
CROCKETT SUBSTATION-LORETO 138 KV Line	Reconductor line	\$2,433,000	12% (\$302,000)
DINOSAUR-LORETTO SUBSTATION 138 KV	Reconductor line	\$2,088,000	69% (\$1,435,000)
SINTEK-ANITA SUBSTATION 138 KV L	Reconductor line	\$993,000	71% (\$706,000)
PLANT 31 PAPER MILL-ANTELOPE MINE SUBSTATION 138 KV	Reconductor line	\$4,094,000	71% (\$2,909,000)
HEALDSBURG-SINTEK 138 KV Line	Reconductor line	\$1,206,000	71% (\$857,000)
ARTESIA COUNTRY CLUB 138/1 kV TX	Replace transformer with higher-rated unit	\$7,151,000	4% (\$312,000)
ARTESIA COUNTRY CLUB 345 kV/1 kV TX	Replace transformer with higher-rated unit	\$15,543,000	4% (\$678,000)
	Total allocated to project		\$20,765,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.